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Quantum Games on Networks

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To my father, who nourished my love for knowledge.

You will be forever in our hearts.

Abstract

This thesis studies the Quantum Prisoner's Dilemma on Networks and the different behaviors of classical and quantum players. We develop a novel formulation for generating quantum circuits for the simulation of quantum prisoner's dilemma games using graph networks from adjacency matrices. We also present a thorough mathematical analysis of the effects for many possible parameterizations in the original model. This analysis explains quantitatively why, almost always, quantum players are in an advantageous position compared to classical players. Moreover, we introduce a technique for classical players to improve their chances against quantum players by keeping their last choice in memory and changing future moves based on it. Finally, we experimentally establish that knowledge sharing not only enriches the classical players' search spaces and strategy options, but also provides a significant enough effect for a classical player to conditionally outperform their quantum counterpart.

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Chapter 1

Introduction

1. Game Theory

1..1 Introduction to Game Theory

In the field of mathematics, Game Theory is the analysis of strategic situations in which there are multiple decision makers (i.e., players) and a decision maker's optimal action can depend on what the other players do[1]. It was first formally proposed by Von Neumann in 1943, where he tried to formulate economic problems using mathematical models into strategy "games" [2]. The principle of Game Theory is the "mathematization" (as referred to by Neumann) of problems is abstraction; By representing a problem in a highly abstracted form, it is possible to not only apply mathematical logic and solutions but also to study a wide range of phenomena using the same or very similar formulations.

A "game" in this context is a description of strategic interaction that includes the constraints on the actions that the players can take and the players' interests, but does not specify the actions that the players do take [3]. This intrinsically abstract definition can be applied to a broad variety of problems by defining the variables such as to conform to each case's specifics. For example, to define the game of chess as a mathematical problem, the players would be the actual players of the game, with the action constraints being the possible moves that are legal in chess, and the interest of either player is to checkmate the opposing king piece. This definition provides the context of the problem, construct the possible search space and establishes the goal of each participant.

1..2 Types of Problems

Depending on the structure and scope of each problem, it can be modulated as either a strategy game or an extensive game [3].

In strategy games, both players are required to make a decision or choose an action independently and without any knowledge of what the other player has chosen. These decisions are made simultaneously, and their results also take effect simultaneously. An example of a simple strategy game is the traditional game of ”Rock, Paper, Scissors”, where the action of a play is a choice between Rock, Paper or Scissors, and the goal to pick the choice that wins over the opponent’s choice. Both players make their choices concurrently, and the results (in this case the reveal of the hand) also take effect together.

Extensive games take into account a new set of conditions. Players may a series of choices, either simultaneously or independently and after the results of the previous actions take effect. In this type of game, the *timing* of an action and the *information* known beforehand by each player play an especially important role. Extensive games can usually be represented as a set of strategy games with each game’s end state functions as the next game’s start state [4]. An auction of an item can be abstracted into an example of an extensive game, where the players are all prospective bidders, each with the goal to buy the auctioned product, and each constrained by their own budget. Each action is taken sequentially, and its result is the change of the item’s price, with ll players making their decisions concurrently, after the new price is decided. Thus, the state of the previous round is the beginning of the current one, and the choices a player will make depend on the round’s state.

1..3 Applications

As Game Theory was developed as a bridge between mathematics and economics, its heavy abstraction can be utilized to formulate and analyze a myriad of problems in several disciples.

Economics

As mentioned above, game theory is a foundational tool in economics used to model strategic interactions among rational agents. In [5] Samuelson provides several examples of game theory being used in economic sciences (among others) to solve the real life problem of calculating a product’s equilibrium price using game theory principles.

Politics

In politics, game theory analyzes strategic behavior in conflicts, voting, negotiations, and coalition formations. In the Cuban Missile Crisis, the need for strategic and temperate actions arose as the risk of a wrong choice could have resulted in

an all-out nuclear war. In [6], Zagare outlines the conflict as a game theory model, and examines how each action could have resulted in a specific outcome.

Biology

Game theory in biology, particularly through evolutionary game theory, is used to understand how behavioral strategies evolve over time as examined in [7]. It models interactions among organisms where payoffs are measured in terms of reproductive success rather than utility or profit. Classic examples include the Hawk-Dove game or strategies in animal mating and foraging behaviors. The concept of Evolutionarily Stable Strategies (ESS) explains how certain behaviors persist in populations due to their stability against alternative strategies.

2. Quantum Computing

2..1 Introduction to Quantum Computing

Quantum Computing is the field of computer science focused on developing computers and algorithms that exploit quantum phenomena to perform massively parallel computations. While classical computers encode data in binary bits (0 or 1), quantum computers use two-state quantum systems to represent quantum bits (or qubits) [8]. Qubits, since they are inherently quantum systems, can exist in a superposition of their states, enabling the simultaneous processing of a vast number of possibilities.

In early 1980, Richard Feynman and David Deutsch laid the theoretical foundation of quantum computing through their works, that proposed that quantum systems could be simulated more efficiently by quantum devices [9], [10]. These early insights led to the development of quantum algorithms in the next few years, most famously Shor's algorithm for integer factorization, which demonstrated a clear computational advantage [11]. In recent decades, rapid advances in quantum hardware and error correction techniques have brought the field closer to practical realization [12].

2..2 Types of Quantum Computers

Quantum computers can be classified into several models based on how they perform computation, including gate-based, topological, and annealing models.

The gate-based model is the most widely studied and resembles classical computing in structure, using quantum gates to manipulate qubits through a sequence

of unitary operations; this model underpins most algorithms designed for general-purpose quantum computation, such as Shor's and Grover's algorithms [13].

In contrast, quantum annealing is a specialized model focused on optimization problems, where the system evolves toward the lowest energy state of a problem-specific Hamiltonian; this approach, used by companies like D-Wave, is not universal but offers practical advantages for certain classes of problems [14].

The topological quantum computing model takes a fundamentally different approach by encoding information in the global properties of anyons manipulated through braiding operations that are intrinsically resistant to decoherence and local noise [15].

2..3 Applications

Quantum computing is an emerging field, and has attracted the attention of many a researcher from a variety of different principles, studying its applications on specific cases. In biology, they may enable more accurate modeling of molecular interactions and protein folding, which could lead to advances in drug discovery and personalized medicine[16].

Quantum algorithms also have shown progress in enhancing machine learning tasks such as clustering, classification, and sampling, where high-dimensional data can be processed more efficiently [17]. An endeavor especially crucial, due to the popularity of LLMs and Machine Learning applications in the recent years.

Lastly, quantum computers are uniquely suited for simulating quantum systems, enabling more accurate modeling of complex molecular interactions, materials, and chemical reactions that are unmanageable for classical computers [18].

3. Quantum Game Theory

3..1 Introduction to Quantum Game Theory

Quantum game theory expands on classical game theory by implementing quantum-mechanics inspired logic into game theory problems, and exploring how it affects the players' decision-making. Instead of fixed or probabilistic strategies, players make choices using quantum states that can exist in several configurations at once. These states can exploit quantum phenomena, like entanglement, linking the players' outcomes in ways that defy classical game theory, or superposition to simultaneously adhere to several states. By treating strategies as physical operations on quantum systems we can simulate quantum game theory games using quantum systems. The quantum operation-rotations, phase shifts and entangling gates become the moves of the game. Because outcomes depend on amplitudes and relative

phases, not just on classical probabilities, what counts as a “best reply” is the combination of all possible payoffs, multiplied by the probability of each payoff. This can lead to quantum games producing payoff changes that are non-intuitive from a classical standpoint. [19]

Stability concepts adapt accordingly. A Quantum Nash Equilibrium is still a profile of strategies from which unilateral deviation is unprofitable, but verifying such a profile requires attention to operator algebra and state evolution rather than to probability vectors alone. Moreover, equilibria found in quantum versions of familiar games often exhibit qualitatively different social properties—greater fairness, higher joint payoffs, or novel mixed–entangled solutions—so the normative conclusions one draws about rational behavior can change once the physical substrate is taken into account. [20]

3..2 Differences with Classical Game Theory

One of the key differences between quantum and classical game theory lies in how strategies and information are represented. In classical games, players select from a finite set of discrete strategies or from probabilistic combinations of said strategies, and the outcomes are fully determined by the payoff matrix. Each decision is independent, and correlations between players’ choices must be established through explicit communication or prior arrangement. In quantum games, by contrast, strategies are implemented as operations on quantum states. Superposition allows a single choice to encode multiple possibilities simultaneously, while entanglement links the outcomes of different players in ways that are impossible classically. This creates a far richer and more flexible strategic environment, where the structure of the game itself can be altered by the underlying quantum properties of the system.

Another major distinction concerns the way interaction and coordination occur. In classical frameworks, players can only influence one another through direct signaling or by observing outcomes over repeated plays. In quantum games, entanglement provides correlations that exist independently of any communication, enabling coordinated results without explicit negotiation. Interference effects further allow certain strategies to reinforce or cancel each other, producing payoffs that cannot be explained purely by probability. These features mean that quantum games can support cooperative or mutually beneficial outcomes even in situations that are competitive by classical standards. In this sense, quantum game theory represents not just an extension of classical reasoning, but a fundamentally different approach to understanding how rational agents make decisions when information and strategy are governed by quantum principles. [21]

3..3 Improvements from quantum logic

Quantum logic offers a new way of thinking about strategic interaction, and one of its most significant effects is the potential for better payoffs. Unlike classical strategies, which are limited to discrete choices or probability distributions, quantum strategies can make use of superposition, allowing a single move to represent multiple possibilities at the same time. This flexibility gives players the ability to explore outcomes that would be inaccessible in a classical setting, and through interference, they can amplify favorable results while diminishing less desirable ones. In situations like the Prisoner's Dilemma, where classical reasoning often leads to suboptimal or purely competitive outcomes, quantum strategies can shift the balance toward cooperation, producing results that benefit all parties involved.

Entanglement further strengthens these advantages by creating correlations between players' choices that do not exist in classical games. By sharing an entangled state, players can align their strategies in ways that increase joint gains or reduce the likelihood of mutually harmful outcomes, even without direct communication. The combination of superposition and entanglement allows quantum strategies to break through classical limitations, enabling outcomes that are both individually rational and collectively more efficient. In effect, quantum logic provides players with tools to navigate complex strategic scenarios in ways that classical methods cannot match, opening new possibilities for achieving higher payoffs and more balanced results. [20, 21, 19]

Chapter 2

Theory

1. Prisoner's Dilemma

The Prisoner's Dilemma is one of the most influential models in game theory, illustrating the conflict between individual rationality and collective welfare. The story is often told through two suspects being interrogated separately. Each suspect is provided with a choice, they can either accuse the other suspect, or remain silent. The rules of the game are known to both players, as are the outcomes. If both suspects remain silent, the interrogators will not have enough evidence to imprison the suspects for the main accusation, but both will be held in prison for a year due to minor charges. If only one suspect accuses the other, and the second remains silent, the latter shall be convicted for 5 years, and the accuser will be let go. Lastly, should both suspects accuse each other, both will be convicted for 3 years. In the context of the prisoner's dilemma, staying silent is usually referred to as "cooperating", and accusing the opponent is called "defecting".

Mathematically, the Prisoner's Dilemma is a two-player game where each player has two possible courses of action. To describe the games potential outcomes, a payoff matrix is created, showing what each player "earns" depending on the combination of choices. In this case, each point corresponds to years in prison, meaning that each player aims for the least possible points every time.

The standard conditions of the game are written as $T > R > P > S$. Where T is the "temptation" payoff for when one player defects while the other cooperates, R is the "reward" for mutual cooperation, P is the "punishment" for mutual defection, and S is the "sucker's payoff for the cooperating player when the opponent defects. These inequalities guarantee that defection is always more attractive individually, even though both players would prefer the cooperative outcome if they could trust each other.

A simple version of the payoff matrix is presented below:

	Cooperate	Defect
Cooperate	(R,R)	(S, T)
Defect	(T, S)	(P,P)

For the example provided above and for all the experiments we will showcase later, the specific payoff matrix is as follows:

	C	D
C	(1,1)	(5, 0)
D	(0, 5)	(3,3)

Table 2.1: Payoff values for every Prisoner's Dilemma combination

1..1 A strategic paradox

In game theory, the most common goal of a player is to either maximize, or, in some instances like the prisoner's dilemma, minimize their total payoff. This state of maximal payoff is called the Nash Equilibrium, and is defined as a situation in which no player has an incentive to unilaterally change their strategy, given the strategy chosen by the other player. Once the equilibrium is reached, each player is making the best possible decision they can, taking into account the decision of their opponent. [22]

In contrast to the Nash Equilibrium, which emphasizes stability of individual choices, another important concept in game theory is Pareto optimality. A Pareto optimal outcome is observed when no player can be made better off without making at least one other player worse off. Importantly, Pareto optimality does not necessarily align with fairness or equality; one player may receive a substantially higher payoff than others, and the outcome can still be Pareto efficient so long as no alternative allocation benefits one player without harming another. [23]

In many games of game theory, it is possible to achieve both Nash equilibria and Pareto optima in the same game state. This, however, is not the case for the Prisoner's Dilemma. In this particular game, the Nash equilibrium corresponds to the outcome in which both players defect. While this outcome is stable in the sense that neither player has an incentive to deviate unilaterally, it is not Pareto efficient, as both players would receive strictly higher payoffs under mutual cooperation. The Pareto-optimal outcome of cooperation, on the other hand, is unstable, since each player retains an incentive to defect in order to improve their individual payoff.

2. Prisoner's Dilemma on Networks

The classical Prisoner's Dilemma isolates a simple tension: individual incentives can run counter to collective welfare. Yet most social interactions do not occur in isolation between two anonymous actors; they occur between several actors, all interconnected with their own unique relationships. When agents play the dilemma repeatedly against a set of neighbors rather than against the same, single opponent, new patterns appear. The networked formulation studies how network structures influence the spread of cooperation or defection in a population where individuals play the game[24].

To formulate the prisoner's dilemma game on a network, a way to calculate the payoff of each player is needed. In this section we will define the prisoner's dilemma game on a network as follows: Let the undirected graph $G = (V, E)$ where each vertex $i \in V$ is an actor, and each edge $(i, j) \in E$. If the possible actions for an actor are $s_i \in \{C, D\}$, and the payoff function of the prisoner's dilemma is $u(s_i, s_j)$, $\{i, j\} \in V$, then the total payoff of actor i is:

$$U_i = \sum_{j \in V} u(s_i, s_j) \tag{2.1}$$

with the usual inequality $T > R > P > S$ for the two-player payoff function $u(s_i, s_j)$.

The topology of G — its degree distribution, clustering, path lengths — matters empirically and mathematically. Regular lattices, small-world graphs, and scale-free networks produce qualitatively different outcomes [24][25] . On lattices, cooperative clusters form and can be resilient because boundary cooperators obtain sufficient reward from their cooperative neighbors to offset the temptation to defect[26]. In contrast, in networks with hubs (high-degree nodes), a defector occupying a hub may exploit many cooperators and thereby spread defection rapidly[27].

In this thesis we will use the network presented by the graph in Figure 2.1 for our experiments. Each vertex in the graph represents a player, and each edge denotes a game of Prisoner's Dilemma between two players. The total payoff of a player is therefore defined as the sum of the outcomes from all games played with its immediate neighbors.

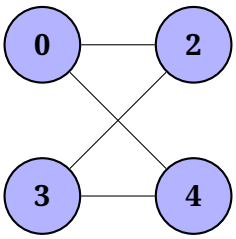


Figure 2.1: A network for the classical Prisoner's Dilemma game with 4 players.

3. Quantum Prisoner's Dilemma

Quantum game theory extends the classical framework by embedding strategic interaction in the language of quantum mechanics. Classical games typically restrict players to a small, discrete set of strategies, but their quantum analogues allow strategies to be represented by quantum operations, greatly enlarging the decision space. This shift introduces phenomena such as superposition and entanglement into the analysis of rational choice. As a result, outcomes that are impossible in the classical setting become accessible, and new solution sets emerge.

Quantum adaptations of classical games broaden the strategic landscape by introducing quantum strategies, which in turn generate new Nash equilibria. The first such formulation was presented by Meyer in the context of a zero-sum game, where the gain of one player corresponds exactly to the loss of the other. Meyer demonstrated that when one player is limited to classical strategies, an opponent with access to quantum strategies can always secure victory[28]. Shortly thereafter, Eisert extended this idea to the well-known Prisoner's Dilemma, a non-zero-sum game. In this influential work, Eisert and colleagues proposed a general framework for two-player quantum games and identified the necessary conditions to ensure that the classical versions are faithfully embedded within the quantum model[**?**]. They also examined scenarios in which a quantum player competes against a classical opponent, showing that a quantum strategy exists which consistently provides the quantum player with an advantage.

This rule extends to the prisoner's dilemma game too, as the interactions in the game can be formulated into the quantum circuit shown in 2.2.

Let players A and B. Each player will be represented as a single qubit, let $|q_A\rangle$ and $|q_B\rangle$ the corresponding qubits. The two players will be entangled using the entanglement gates E and $E^\dagger$, and each player will a one-qubit quantum gate to select their strategies from gates U_A and U_B respectively. The measurement of end state of the circuit will be considered as the final game state, and the payoffs of each player will be calculated accordingly.

Thus, the initial state of the game will be

$$|q_{in}\rangle = |q_A q_B\rangle \tag{2.2}$$

and the end state will be

$$|q_{out}\rangle = E^\dagger (U_A \bigotimes U_B) E \, |q_A q_B\rangle \tag{2.3}$$

-=

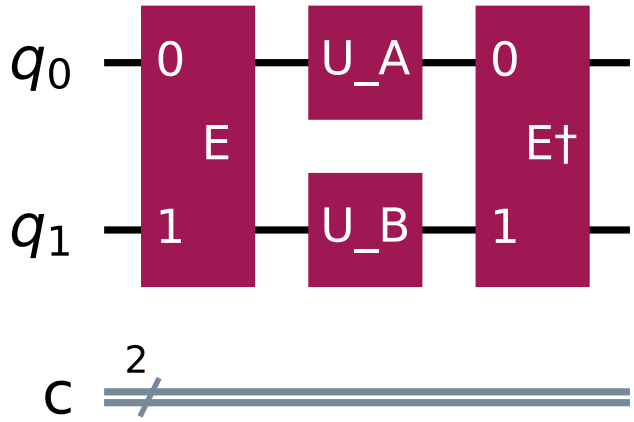


Figure 2.2: A two player quantum circuit for the Prisoner's Dilemma

4. Quantum Prisoner's Dilemma on Networks

The quantum extension of the Prisoner's Dilemma replaces classical binary choices with operations on quantum bits and uses entanglement and quantum gates to expand the available strategies and states an actor may take. This, coupled with the game's expansion on networks, where both the graph structure and topology affect all actors, can provide a highly expansive search space where new player behaviors may be observed. However, research of the application fo the quantum prisoner's dilemma on networks has been very limited. Understanding how quantum players interact with network structure is crucial to uncovering the conditions under which cooperation can emerge and stabilize, especially in comparison to classical networked games.

In [29], Li et al. investigates the dynamics of quantum strategies within evolutionary games played on small-world networks. Actors are assigned quantum strategies from a predefined set and engage in multiplayer quantum games with their neighbors. Quantum strategies are observed to evolve over time, and appear to dominate populations initially dominated by classical strategies. Quantum strategies can successfully invade populations even when defectors are prevalent, and they always outperform classical strategies in terms of cooperation levels.

In [30], the dynamics of quantum strategies in multi-player evolutionary games are explored, specifically focusing on a three-player version of the Prisoner's Dilemma. Hypergraphs are implemented to model the simultaneous interactions among three

agents, where each hyperedge connects exactly three nodes, representing the players. They investigate two types of hypergraph networks: a random network and a scale-free (SF) network, to examine how network structure influences the evolution of quantum strategies. In certain scenarios, defection strategies can perform as well as quantum strategies, highlighting the complex interplay between quantum strategies and network topology in multi-player evolutionary games.

Most of the bibliography focuses on multiplayer quantum prisoner's dilemmas, using graph networks to signify the multiple players participating in the game. In this thesis, we will introduce a new formulation of quantum prisoner's dilemma on networks, using a graph G to represent the pairs of players that will be participating as opponents in a series of 2-player games, indicated by graph G.

Chapter 3

Experiments

1. A circuit for the quantum prisoner's dilemma

1..1 A quantum circuit for the classical prisoner's dilemma

In this chapter is we present the steps for the construction of a quantum circuit that models the Prisoner's Dilemma in order to be played using quantum computers. The quantum circuit for a two player game of the Prisoner's Dilemma is constructed using a pair of qubits, each representing a single player. The initial state of the system is prepared in the computational basis $|00\rangle$, which we arbitrarily set to correspond to both players initially intending to cooperate. We can represent the classical Prisoner's Dilemma strategies (opting to either Cooperate and Defect) as gates in the quantum circuit. As state $|00\rangle$ was selected to represent cooperation, the *action* of cooperating shall be represented by the quantum identity gate I, while *action* of defecting will be represented as the NOT quantum gate.

For the classical Prisoner's Dilemma game we can create a quantum circuit that maps each combination of player states to the Prisoner's Dilemma's payoff matrix. Thus, we get the following payoffs per state:

$$\begin{bmatrix} |00\rangle \rightarrow (1,1) & |01\rangle \rightarrow (0,5) \\ |10\rangle \rightarrow (5,0) & |11\rangle \rightarrow (1,1) \end{bmatrix} \tag{3.1}$$

Thus a quantum circuit for the two player classical Prisoner's Dilemma is as follows:

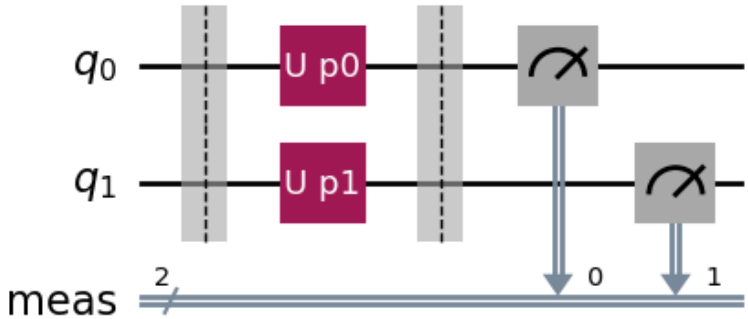


Figure 3.1: Quantum circuit for the two player classical Prisoner's Dilemma

1..2 A quantum circuit for the quantum prisoner's dilemma

To simulate a prior knowledge of each other, an entangling operation is applied via the combination of a Hadamard gate on the first qubit followed by a controlled-NOT (CNOT) gate with the first qubits as its source, and the second as its target, producing the Bell-type entangled state:

$$|\psi\rangle = 1/\sqrt{2}(|00\rangle + |11\rangle) \tag{3.2}$$

Following entanglement, each player applies the parametrized unitary operation $U_i(\theta_i, \psi_i)$ to their respective qubit to encode their chosen strategy. This unitary is intentionally left as a parametrized gate, as any single-qubit quantum can be used interchangeably, by calibrating the corresponding parameters. In this case, when $\phi = \theta = 0$, the Unitary gate acts like an Identity gate.

$$U_i(\theta_i, \phi_i) = \begin{bmatrix} \cos(\frac{\theta}{2}) & -e^{i} \sin(\frac{\theta}{2}) \\ e^{i\phi} \sin(\frac{\theta}{2}) & e^{i\phi} \cos(\frac{\theta}{2}) \end{bmatrix} \tag{3.3}$$

After the strategic operations, a disentangling procedure, the inverse of the initial entangling step, is applied to map the quantum state back onto the computational basis for measurement. The final measurement of both qubits yields the classical outcomes of the game, which are then used to compute the players' payoffs according to the original Prisoner's Dilemma payoff matrix shown in 1..

A complete quantum circuit for the Quantum Prisoner's Dilemma for two players is presented below.

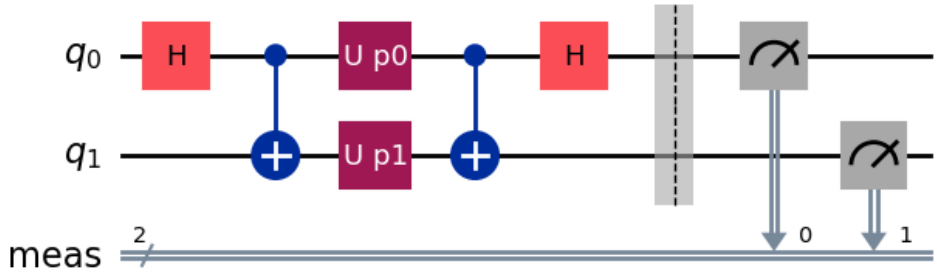


Figure 3.2: A quantum circuit for a two player Quantum Prisoner's Dilemma

To calculate the quantum operation for the quantum Prisoner's Dilemma, we need to simply calculate the end state of the quantum circuit. For ease of calculations, we shall consider the entanglement gates (The combination of Hadamard and $CNOT$ gates) as a single unitary operator $\hat{E}$. Conversely, the disentanglement operator will be depicted as the adjoint of $\hat{E}$, $\hat{E}^{\dagger}$. Thus, the function to calculate the end state $|q\rangle$ of the circuit becomes:

$$|q\rangle = \hat{E}^{\dagger}(U_{p0} \bigotimes U_{p1})\hat{E}\,|00\rangle \tag{3.4}$$

As we mentioned in 1..1, if $\hat{E} = I = \hat{E}^{\dagger}$, and $U \in \{I, X\}$ then the circuit represents a classical Prisoner's Dilemma Game

2. Mapping the network to the circuit

In section 2. we showcased the formulation of the Prisoner's Dilemma using network graphs. In this thesis we expand this application to propose a formulation for the Quantum Prisoner's Dilemma using network graphs. In our formulation we develop a simple, bidirectional network graph with 4 nodes and 2 edges on each node. In this graph, players are represented by the nodes, and the games played between two players are denoted by the edge between them. For a more illustrative example, in graph 2.1 node 0 is connected with nodes 1 and 3, leading to two games for player 0, against players 1 and 3. For ease of recreating experiments the order of the games remains constant, following the order of the nodes. Player 0 always plays all their games first, then player 1 and so forth, until no players or games remain.

For graph 2.1, the order of the games would be:

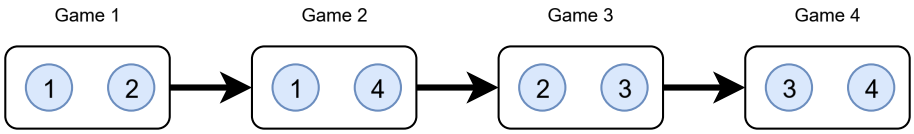


Figure 3.3: Game order for network graph 2.1

Using the above game order, we can now construct a quantum circuit of consecutive 2 player Prisoner's Dilemma to form an extensive game consisting of multiple strategy games, as shown in Figure 3.4. It is important to note that a single qubit in our circuit does not correspond directly to a single player and vice-versa. Instead, each qubit corresponds to an instance of a player playing a game; a player that would play in two separate games will occupy a different qubit in each game. This brings the number n of qubits needed for the circuit to $n = 2 * |E|$, where $|E|$ the number of edges in graph G.

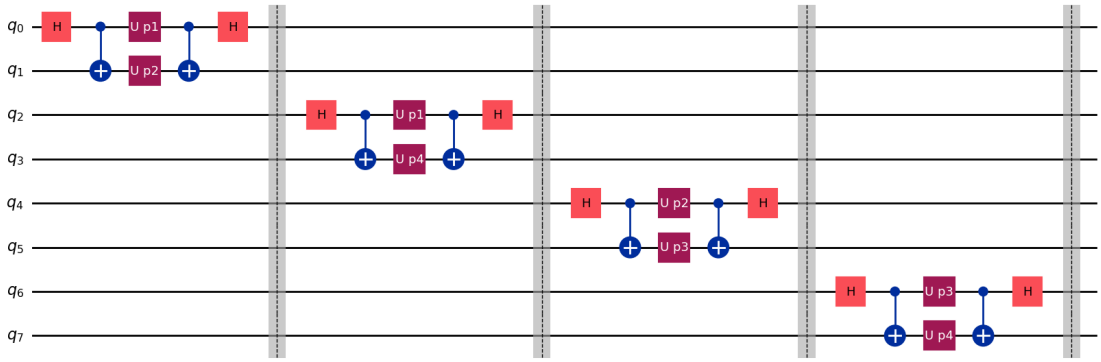


Figure 3.4: A quantum circuit for the game order 3.3

By formulating the network as a series of 2 player games in a quantum circuit, we can now introduce the quantum player to the game. From now on, when a network graph include a quantum player, the player shall be denoted by a red node, to differentiate it from the classical players, as shown in 3.5. In the quantum circuit there is no visual distinction of the quantum player, however the Unitary gate U for the strategy of the quantum player is parametrized to represent a Hadamard gate, instead of the classical players' I or X gates.

Now, using the notation as in 3.4, the state of our quantum register becomes:

$$|q\rangle = \hat{E}^{\dagger}(U_{p_{n_1}} \bigotimes U_{p_{n_2}})\hat{E}\,|00000000\rangle\,\forall n \in g = \{\{0,1\},\{0,3\},\{1,2\},\{2,3\}\} \tag{3.5}$$

Where g is a set of sets containing the edges if graph G.

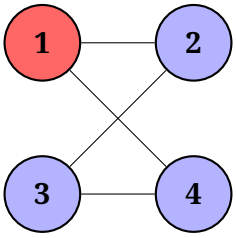


Figure 3.5: A network for the Prisoner's Dilemma game with 1 quantum player. Blue nodes are classical players, red one is the quantum player.

As the index of each node indicates the order in which the node's player will participate in a game, the position of the quantum player is an important characteristic of graph G to keep track of. By changing the position of the quantum player to each of the available indices, 4 distinct graphs are created, all of which will have the quantum player in a different order of play as shown in Figure 3.6.

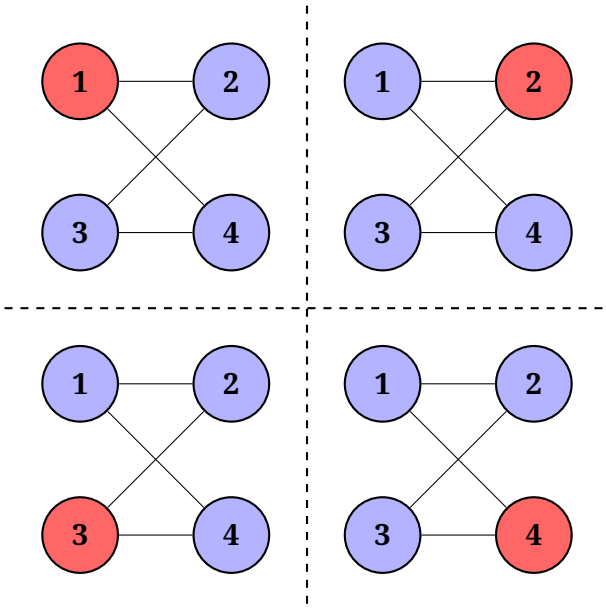


Figure 3.6: All possible positions for the quantum player, shown in the red nodes.

3. Knowledge sharing

In most implementations of Prisoner's Dilemma games, all players have a way of changing their strategy over time. In our implementation we opted for using a knowledge sharing feature, to allow players to pass limited knowledge to themselves, in the next game they play. This was accomplished by adding a $CNOT$ gate on each player's qubit after every game. The source of each $CNOT$ gate is the spe-

cific player, and the target is the qubit that corresponds to the same player, in the next game in order.

Figure 3.7 showcases a quantum circuit for Graph 2.1 with knowledge sharing enabled. Each strategy gate U displays the corresponding player, and every time a player would play a new game, the $CNOT$ gate transfers the output state of the player to become the input state of their next game. Thus, player will present a behavior similar to ”remembering” their previous choice, and they are inclined to repeat it.

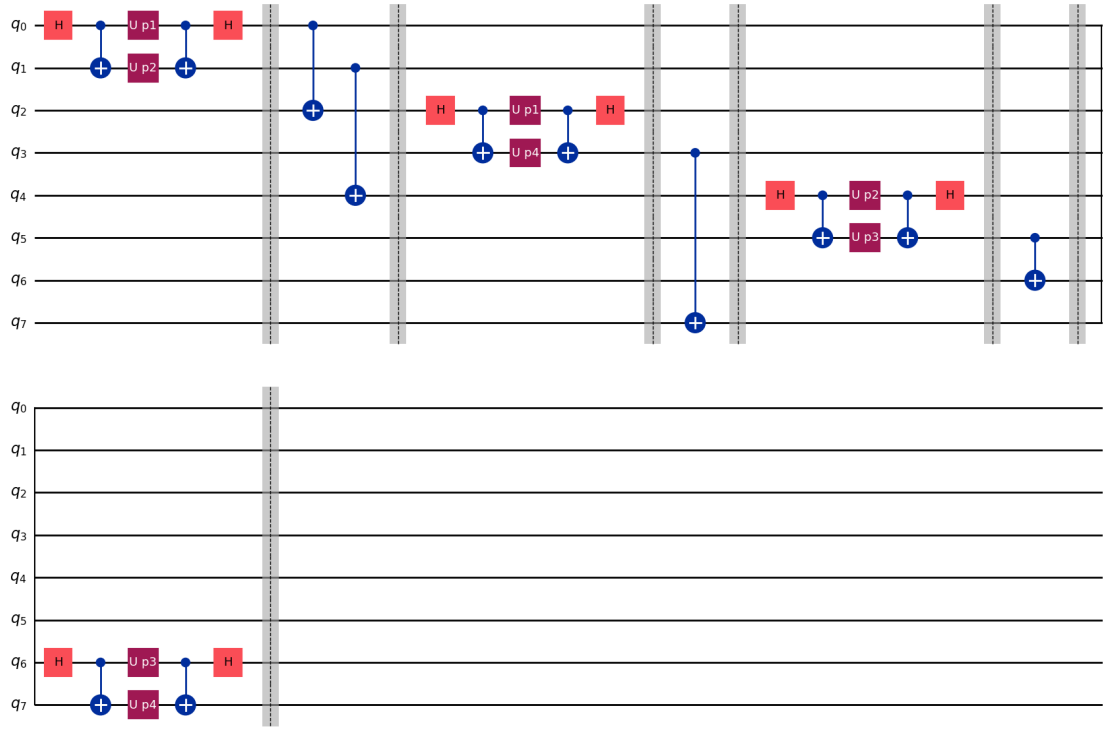


Figure 3.7: A quantum circuit for the game order 3.3 with knowledge sharing

By defining the 2 player quantum game as a 2-qubit quantum operation $D = \hat{E}^{\dagger}(U_{p1} \otimes U_{p2})\hat{E}$, the operations in the circuit are:

$$|q\rangle = (CNOT_{m_n \to m_{n+1}} \bigotimes I)(D_{(p_n, p_{n+1})}) \, |00000000\rangle \tag{3.6}$$

Where g is a set of sets containing the edges if graph G, and m the set of sets containing the position of the next qubit that corresponds to each player's next game. Using graph G, each set is as follows:

$$g = \{\{1,2\},\{1,4\},\{2,3\},\{3,4\}\}, \quad m = \{\{0,2\},\{1,4\},\{5,6\},\{3,7\}\} \tag{3.7}$$

Note that the elements of g are the *players* of each game, while the elements of m are the specific qubits that need to be connected. Thus, it is evident that in order

to construct a quantum circuit for a bidirectional graph G with n edges, $2 * n$ qubits are required.

Chapter 4

Results and Analysis

1. Quantum Advantage?

As explained in section 4., a quantum player with access to quantum strategies is expected to have an advantage over classical players, as they are capable of electing for strategies that can achieve Nash and Pareto equilibria simultaneously. Using graph 2.1 we conducted several experiments to validate the existence of said advantage, as well as to analyze how different specifications and modifications of the circuit and graph G affect the efficiency of the quantum player, and the outcome of the game as a whole.

The first experiment to determine the performance of the quantum player was to execute the circuit and gather the average payoffs of each player. Each experiment of an execution for the circuit depicted in 3.7 with 2048 shots to the quantum (or simulated quantum) backend. For each game, the payoffs are the same as in matrix 2.1, and as they represent years in prison, the goal of each player is to *minimize* their total payoffs. For every experiment, the "Payoff" of each player is calculated as each state's payoff multiplied by the probability of the specified state. The payoffs are averages when a player plays in multiple games, and each payoff is the sum of the payoffs for both rounds of play in each game.

1..1 Experiments on Aer Simulator

Initially, we ran the experiments using IBM's Aer Quantum Simulator Backend [31] to simulate the execution of the circuit in a quantum computer. In Figure 4.1 we showcase the average payoffs of all players in the game. This visualization shows that the quantum player (in our case, according to graph 2.1 the quantum player is Player 1), does indeed perform relatively better than most players, except for player 4, an interesting finding which we will expand on in the coming sections.

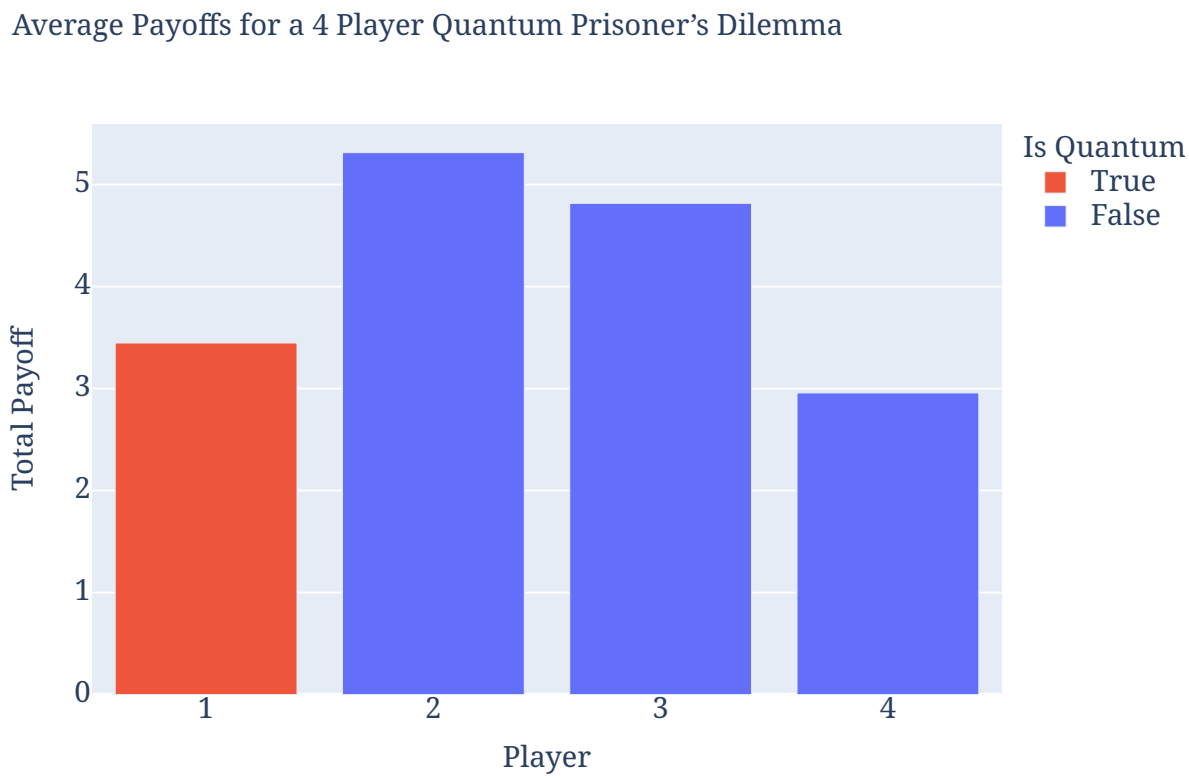


Figure 4.1: Average payoffs of every player (less is better)

To analyze the first results deeper we showcase the distribution of each player's payoffs using the boxplot shown in Figure 4.2. It is evident that while the median value of all players was 1, which according to 2.1 can only occur when both players decide to cooperate, players 1 and 2 are more likely to get higher (worse) payoffs than players 3 and 4. By inspecting the game order in Figure 3.3, players 1 and 2 play in the earlier games of the round, while player 3 and 4 in the latest.

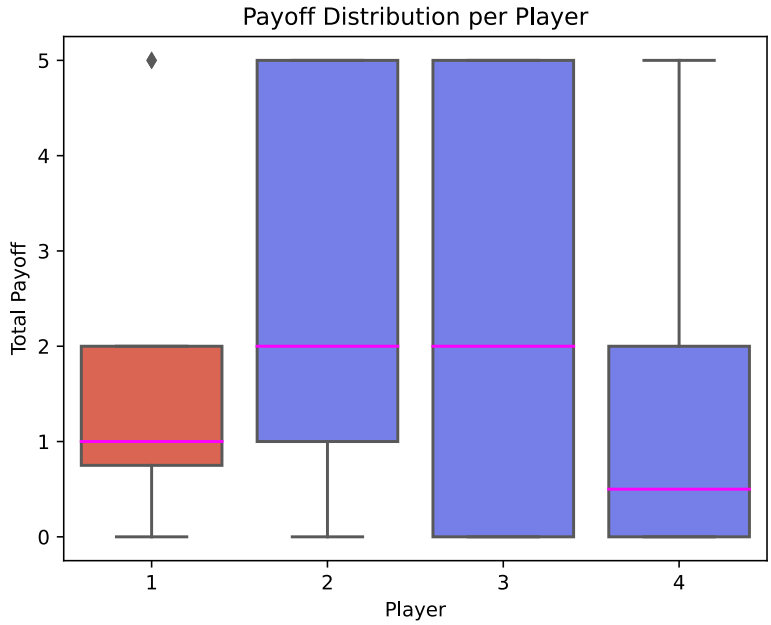
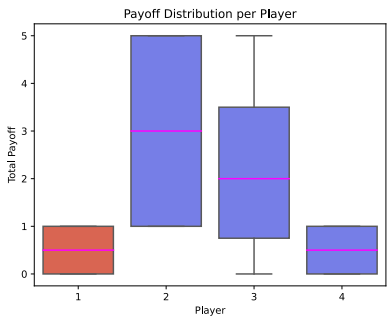
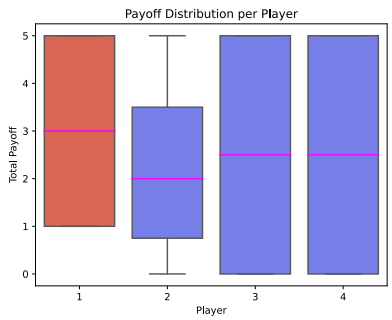


Figure 4.2: Distribution of Payoffs in each player

Another important fact to take note of is that in every game each player participates in 2 rounds, and shares knowledge with oneself about the previous round. This leads to vastly different results for most players in each round, as shown in Figure 4.3. On the first round of games, the quantum player performs significantly better than the classical ones. However, this seems to change in the second round, where the quantum player's median payoff is 5, the worst possible outcome. Despite the drastic change in performance on the second round, the quantum player reaching an the second lowest average showcases an advantage over classical players.



(a) Round 1



(b) Round 2

Figure 4.3: Payoff Distribution per Player by Round

1..2 Experiments on QPU

For a more complete overview of the circuit's behavior, we ran the same set of experiments using IBM's QPU systems [32]. In Figure 4.8 we showcase the average payoffs of all players in the game. Similar to when running the circuit on the Aer Simulator, the quantum player does provide slightly payoffs near the optimal value, without significantly outperforming the other players. The slight variation seen on the average payoffs, especially for players 2 and 3, between Figure 4.1 and Figure 4.4 can be attributed to noise generated by the QPU, and to the stochastic nature of the quantum circuit.

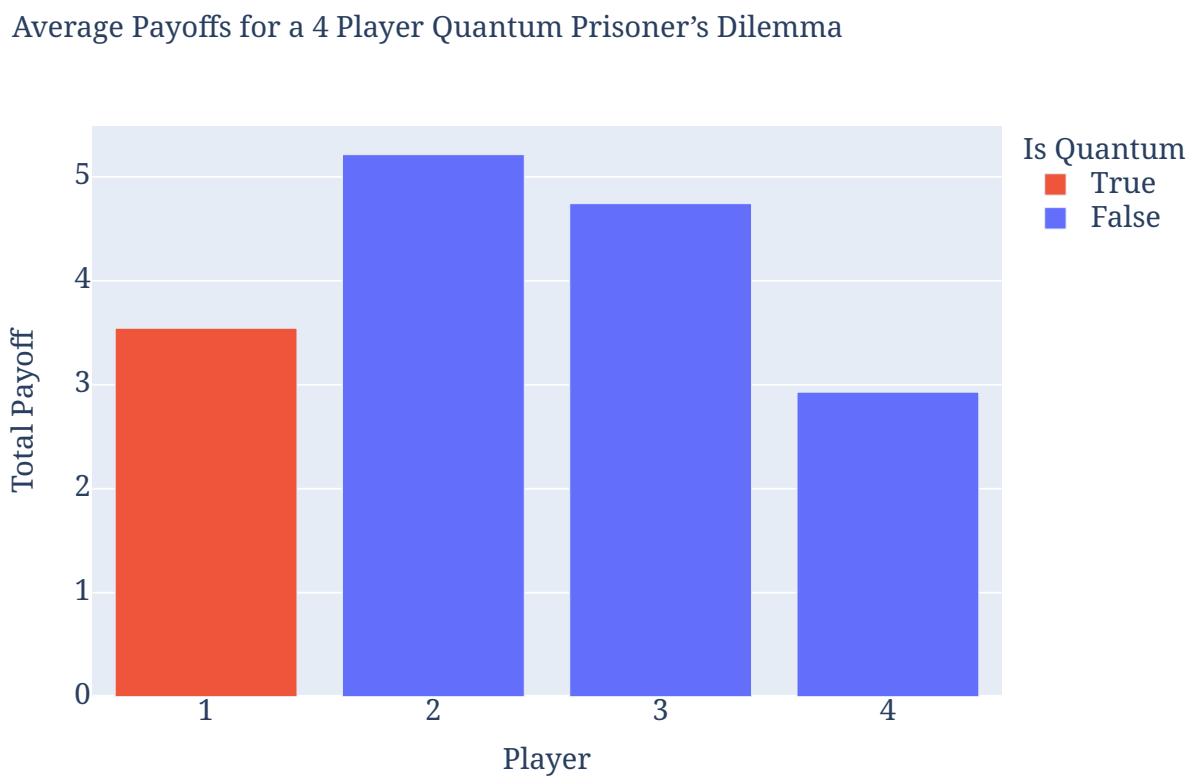


Figure 4.4: Average payoffs of every player (less is better)

2. Effects of player positioning

In Figure 3.6 we highlighted the possible variations of graph G depending the quantum player's index. This, coupled with the improved performance witnessed on players that participate in later rounds, and the fact that all classical players use

identical strategies, signifies that the position of a player may affect their performance. To examine this notion, we ran a new set of experiments for all possible positions of the quantum player.

2..1 Experiments on Aer Simulator

Figure 4.5 shows the average payoff per player when playing as a classical player, as well as a quantum one. Here the improvement of a quantum player's performance is even more perceptible. Every player showcases far better results when playing using quantum strategies, and provides an explanation for the case in Section 1..1, where the quantum player wasn't outperforming all players. There is a clear benefit for the players when playing in the last games, and when combined with utilizing quantum strategies players are even able to reach an average of exactly 1. Another significant insight is that the influence of the player position to the payoffs is so great that it may outweigh the results' improvements by the use of quantum strategies, as the average of the quantum counterparts of players 1 and 2 are almost the same as the results for the classical players 3 and 4. The reason of this discrepancy that the graph layout invokes has not yet been explained, and will be further analyzed in the upcoming sections.

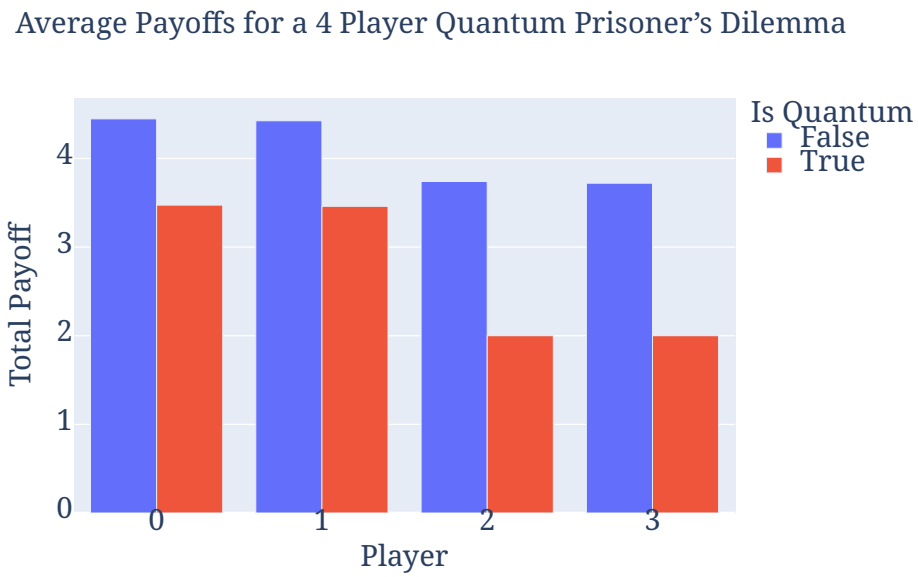


Figure 4.5: Average Payoff per Player and Quantum Player Position

Similarly to Section 1..1, Figure 4.6 presents the distributions of the payoffs for all players and for all quantum player's positions. Again, player 1 and player 2 showcase worse results than players 3 and 4, further highlighting the weight that

the graph's topology has on the results of a game. When players 4 and 5 employ quantum strategies, they exhibit a median payoff of 2, meaning they not only opt for cooperation far more often, but in later games cooperation is more likely to emerge.

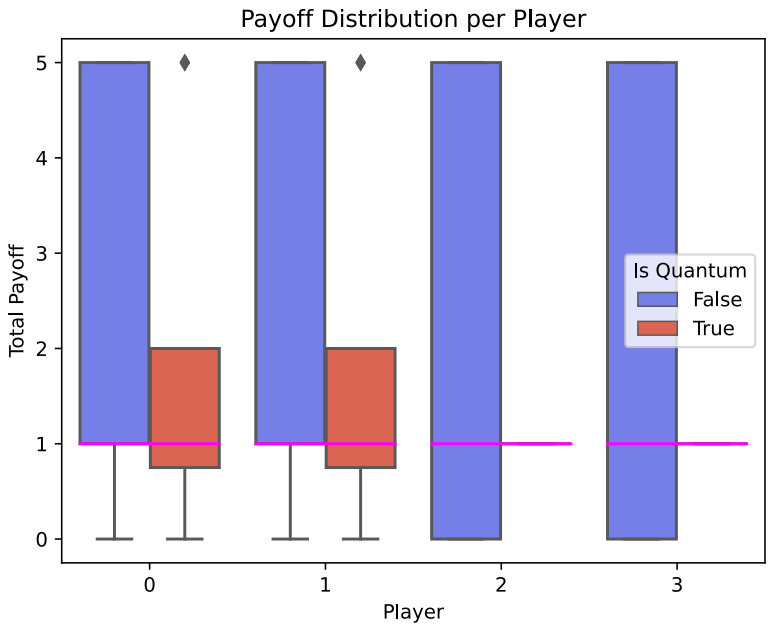
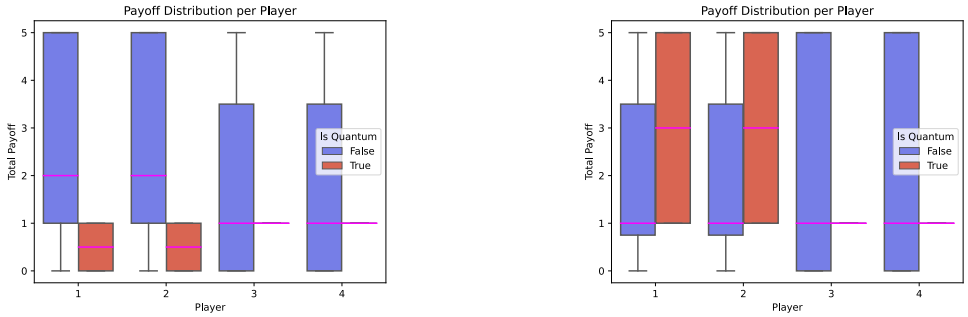


Figure 4.6: Payoff per Player and Quantum Player Position

A deeper investigation on quantum players' performance

To unravel cause of the worse results of quantum players 1 and 2 compared to the quantum players 3 and 4 we need to investigate the results further. Again, by differentiating between the first round a player plays and the second, we are presented with a much clearer image of the inner workings of the game.



(a) Round 1

(b) Round 2

Figure 4.7: Payoff Distribution per Player by Round

Figure 4.7 clearly showcases the discrepancy between the results of players 1 and 2 for their first and second round, as well as the differences for all classical players' payoffs between the rounds. While at first glance the notion of the quantum player performing significantly worse than the classical counterpart seems antithetical to the established literature, this behavior is in fact expected. In our implementation, the feature of knowledge sharing is introduced as a way to pass information from one player's first round to the second. To achieve this, a $CNOT$ gate is used, passing the current state of the qubit onto the next qubit that represent the player.

For a more illustrative example, in the circuit shown in Figure 3.6, qubits $q0$ and $q1$ represent a game between players 1 and 2 respectively, with player 1 having access to quantum strategies while player 2 can only use classical ones. Thus, gate U_{p1} acts as a Hadamard gate, while gate U_{p2} acts as an Identity gate. After the game is finalized, player 1 passes knowledge to their next game using the $CNOT$ gate from $q0$ to $q2$, and subsequently passing their superposition to qubit $q2$.

The next game is between players 1 and 4, represented by $q2$ and $q3$ respectively. During the entanglement of $q2$ and $q3$, $q3$ also goes into superposition. Gates U_{p1} and U_{p4} now act on qubits in superposition, with gate U_{p1} acting again as a Hadamard, and gate U_{p4} as an Identity. However, in contrast to the first game, the input state of both qubits is a superposition state hence gate U_{p1} acting on player 1 cancels their superposition, while gate U_{p4} lets player 4 keep their state. This forces the quantum player 1 to transition into a classical player, while player 4 becomes a quantum player.

2..2 Experiments on QPU

The clear advantage of the quantum player showcased in Figures 4.5 and 4.6 is again present in the experiments conducted using real quantum systems. In Figure 4.8 the average payoffs of the quantum players consistently outperform their

classical counterparts, and exhibit improved results than when executing the circuit on the simulator. The payoffs' distribution obtained from the QPU remains stable across repeated runs, indicating that the observed advantage is not a statistical improbability. These results confirm that the quantum player maintains a measurable performance benefit under all conditions, supporting the consistency of the findings across both simulated and physical implementations.

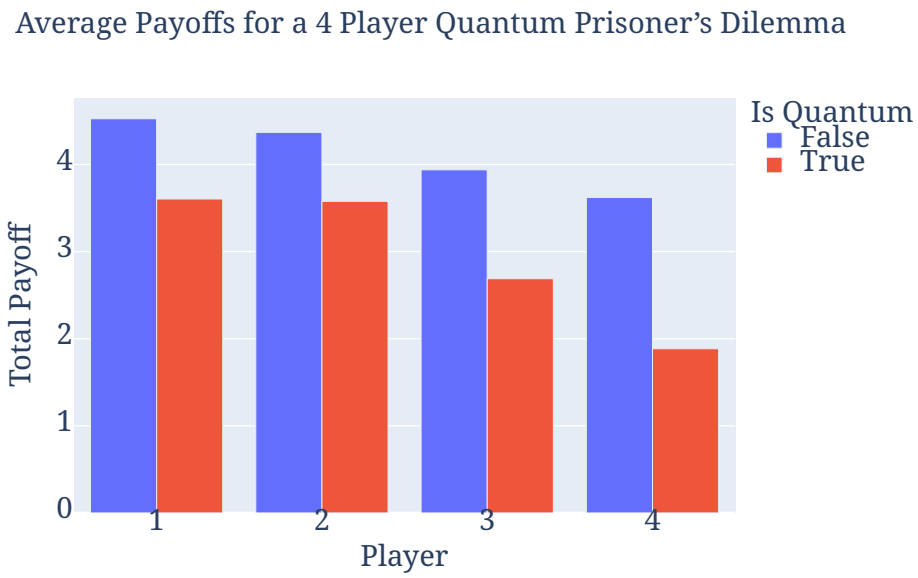


Figure 4.8: Average Payoff per Player and Quantum Player Position

A key distinction between the performance in the QPU and Aer Simulator is seen in Figure 4.9. The distribution of all quantum players payoffs has an extremely high variance, while maintaining a median value of 1. This indicates that quantum players are far more likely to achieve cooperation with another player.

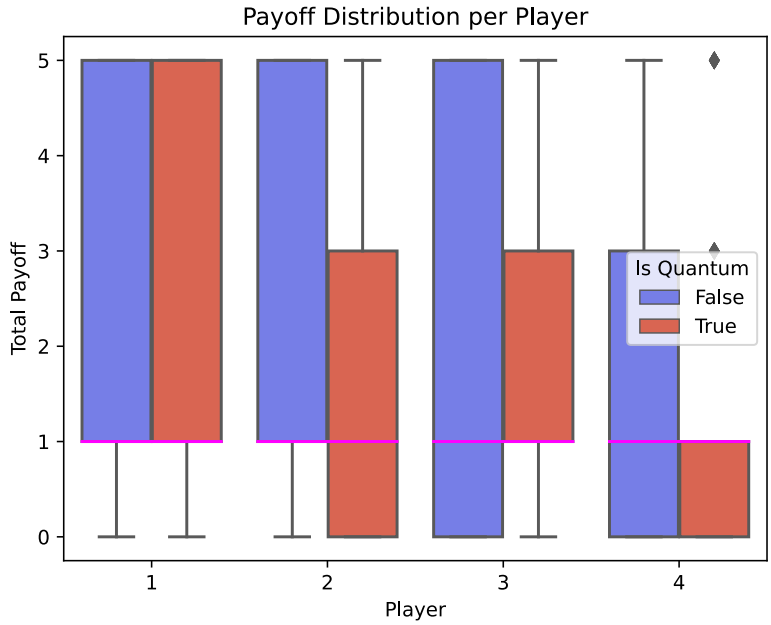


Figure 4.9: Payoff per Player and Quantum Player Position

3. Effects of knowledge sharing

In section 3. we explained knowledge sharing as a way for the players in the game to ”remember” their past choices. This behavior enables players to potentially alternate strategies between rounds and create more dynamic search spaces. This chapter presents the results of the experimental investigation into the effects of knowledge sharing, with the primary objective of the experiment being to assess how enabling or disabling this feature influences the results of the quantum player against the classical ones.

The key point of our analysis is to estimate the extend in which knowledge sharing affects the quantum player. Thus, we experimented by executing experiments comparing games with knowledge sharing logic enabled, and games without it.

3..1 Experiments on Aer Simulator

Figure 4.10 shows the average payoff per player for both their classical and quantum iterations in a game without any knowledge sharing. Without knowledge sharing classical players never change strategies and always opt for Cooperation, as this is has been selected as the default classical strategy. The absence of different strate-

gies eradicates the benefit players get from positioning, and every player position (and every game) produces almost the same average payoff. Quantum players do exhibit a distinct advantage, having an average payoff of 1 for all positions, as graph topology is no longer a factor in the circuit.

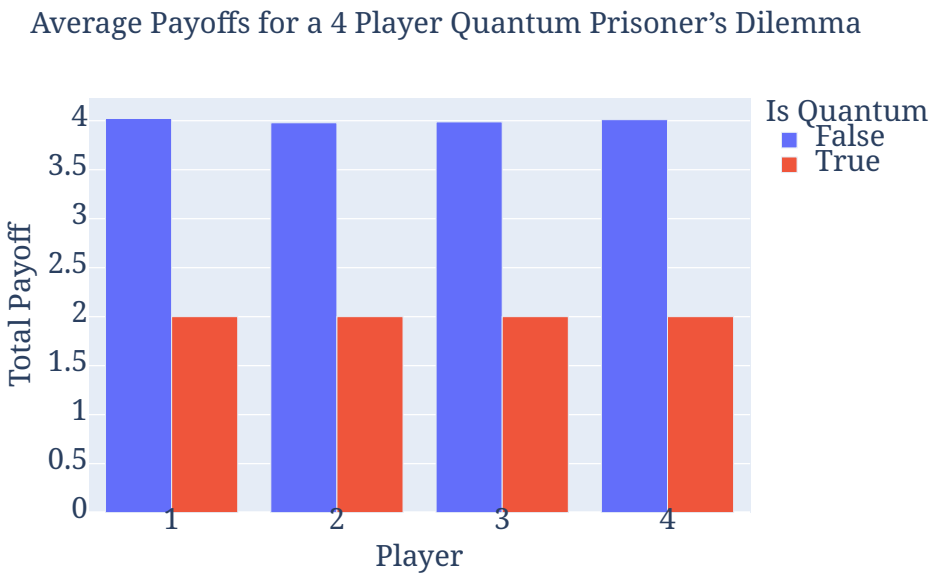


Figure 4.10: Average Payoff per Player without Knowledge Sharing

The payoff distribution of each player is also similarly affected, with all players exhibiting a similar distribution, where classical players may have payoffs of all values. By contrast, the quantum players demonstrate a distribution that aligns in its central tendency, with all quantum players showcasing a payoff of exactly 1. An interesting insight that emerges from Figure 4.11 is the convergence of all players, regardless of their strategic framework, toward a common median payoff of 1. This median value is significant: it points to a systemic inclination toward cooperative outcomes rather than competitive or defection-dominant dynamics.

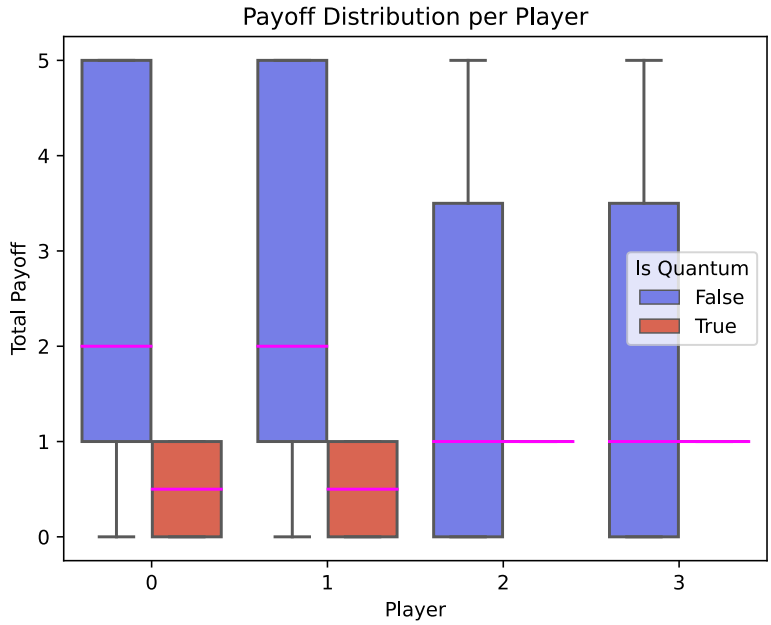


Figure 4.11: Average Payoff per Player without Knowledge Sharing

3..2 Experiments on QPU

Running the experiments on IBM's Quantum Processing Unit (QPU) gives us a clearer picture of how the strategies hold up outside of simulation. The quantum players still average a payoff of 1, as shown in Figure 4.12, which shows that their advantage remains even without any knowledge sharing between games. The classical players, on the other hand, show more fluctuation in their results compared to the clean simulator output. This variation is expected, since real quantum hardware introduces noise and small errors that the simulator does not capture. Even with that variability, the overall trend is the same: classical players earn lower payoffs when knowledge sharing is removed. Without the $CNOT$ gate adjusting their state beforehand, they fall back on their fixed strategy of Cooperation, which limits their flexibility and keeps their average payoff lower than that of the quantum players. In short, the experiments confirm that quantum strategies are robust to hardware imperfections, while classical strategies remain disadvantaged when they cannot make use of knowledge sharing.

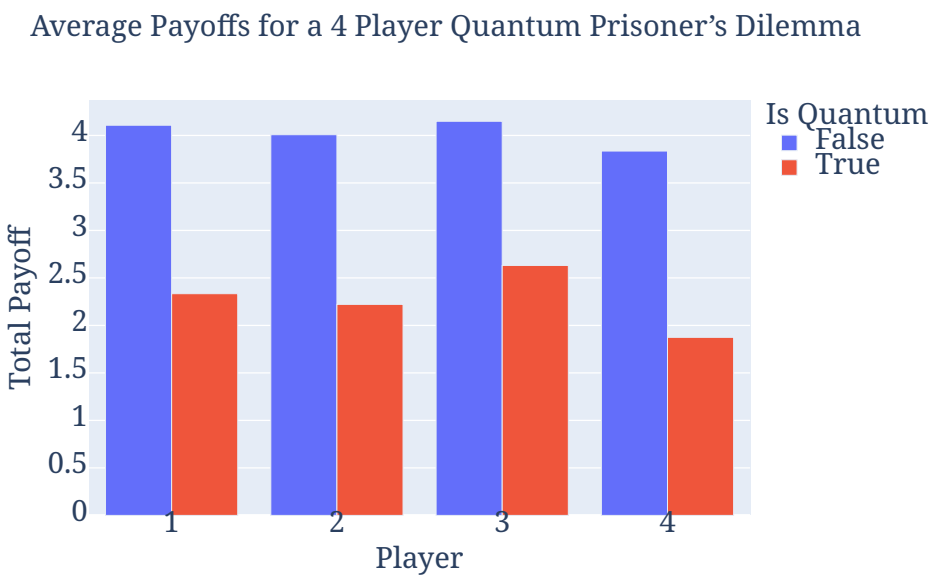


Figure 4.12: Average Payoff per Player without Knowledge Sharing

The payoff distribution shown in Figure 4.13 provides a clearer view of how both classical and quantum players perform on real hardware. For the quantum players, the distribution is tightly centered around a median payoff of 1, reinforcing their tendency toward cooperation. The near absence of deviation suggests that this outcome is not an exceptional occurrence but a consistent feature of their strategy in QPU-based experiments. Classical players, in contrast, display much greater spread in their payoffs than what was observed in the simulator runs. This added variability is consistent with expectations, since quantum noise and hardware imperfections naturally affect measurements on a real QPU. Even so, the overall picture remains the same: quantum players reliably secure cooperative outcomes, while classical players face more uncertainty in their results when the system is implemented experimentally.

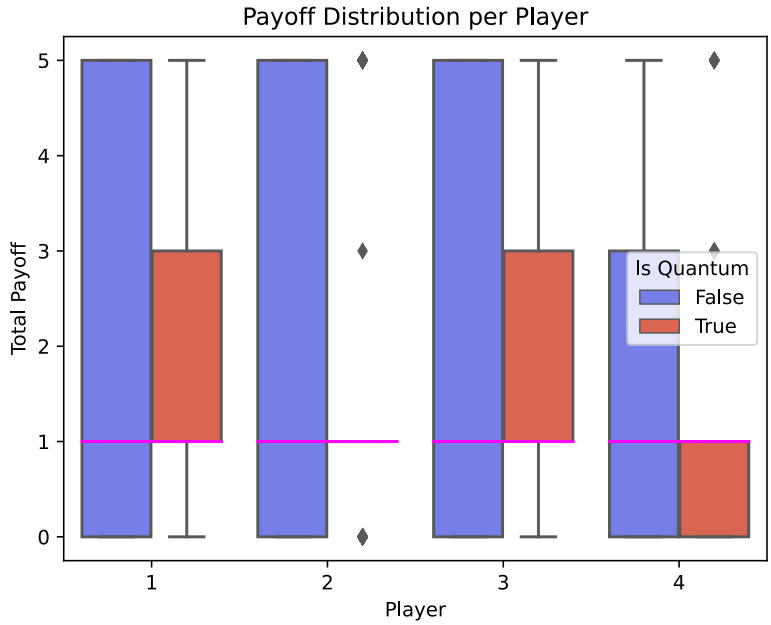


Figure 4.13: Average Payoff per Player without Knowledge Sharing

4. Effects of different knowledge sharing techniques

As shown in Section 3., the existence of knowledge sharing is an important factor in our circuit, as it provides classical players with an opportunity to select a different strategy from their default one. Additionally, this feature is the reason for the findings in Section 2., that the player positioning in the graph topology affects their performance, specifically by providing players that play last with a significant advantage.

To further analyze the effect of the knowledge sharing feature we experimented by using the Controlled Quantum Hadamard Gate (CH) 4.1. The CH gate will significantly change the behavior of our classical players, as they will no longer be strictly confined to classical strategies. The CH gate applies the Hadamard gate on the target qubit instead of the NOT gate. Thus, classical players may become quantum ones, at least for the second round that they will participate in.

$$CHq_0, q1 = I \bigotimes |0\rangle \langle 0| + H |1\rangle \langle 1| = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 & 1 & 1 \\ 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{pmatrix} \tag{4.1}$$

4..1 Experiments on Aer Simulator

Figure 4.14 shows the average payoff per player for both their classical and quantum iterations in a game with CH as the knowledge sharing technique. In this case, the advantage that the later players gain is even more evident, as even when players 2 and 3 begin the game using classical strategies, they exhibit improved results compared to 1 and 2. Quantum players in the first positions also gain a significant improvement, as their averages are much lower than when using the $CNOT$ gate for knowledge sharing. This is due to

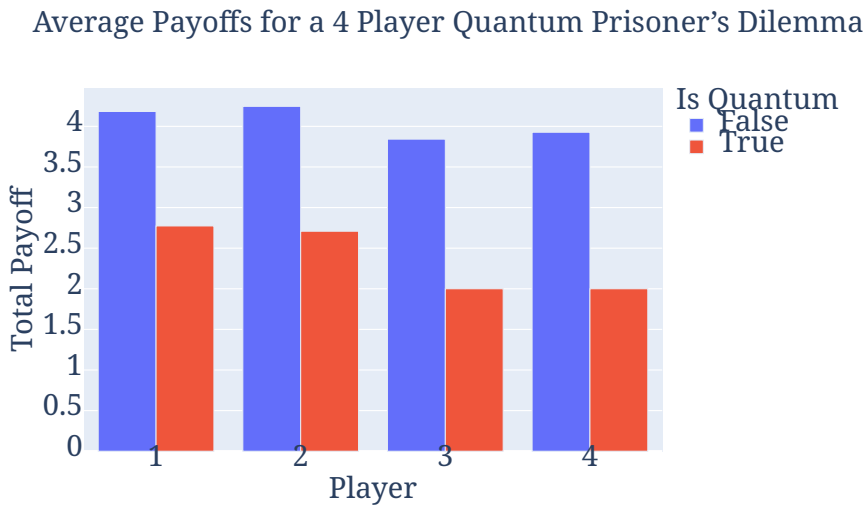


Figure 4.14: Average Payoff per Player for Quantum Knowledge Sharing using CH

The payoff distribution of each player is also similarly affected, with classical players having higly varied results, while quantum players being more concentrated with few outliers, specifically for players 1 and 2. An exceptionally interesting insight however is that all players, classical and quantum, possess a median value of 1, denoting an highly increased tendency for Cooperation across both types of players.

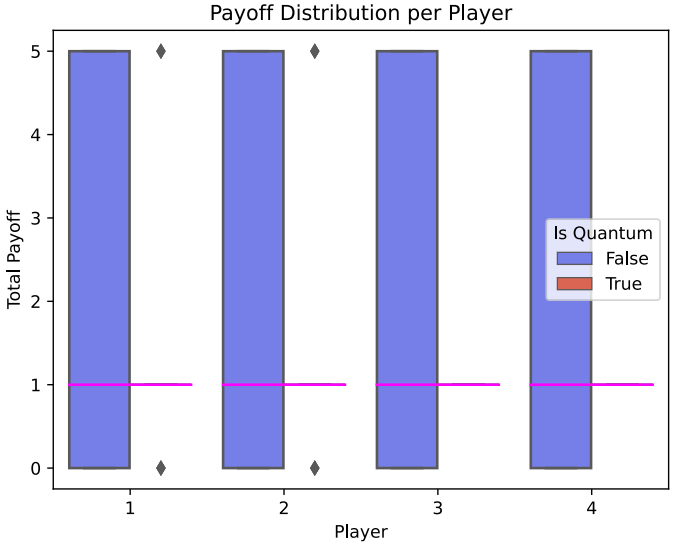


Figure 4.15: Payoff per Player and Quantum Player Position

4..2 Experiments on QPU

Similarly to the experiments conducted with the Aer Simulator, classical players who participate in earlier games tend to perform worse on average, while those appearing in later games achieve noticeably stronger results, with player 3 coming very close to a total payoff of 2, as seen in Figure 4.16. This pattern aligns with expectations: players in later rounds are will always encounter at least one game where both participants can access quantum strategies, thanks to the influence of the CH gate. Under these conditions, the emergence of collective cooperation becomes significantly more likely. Figure 4.17 showcases the effects of this phenomenon much more clearly. Quantum players, by contrast, continue to hold a clear advantage. Their results remain consistently strong regardless of position, showing that their ability to reach optimal outcomes is unaffected by turn order or game placement.

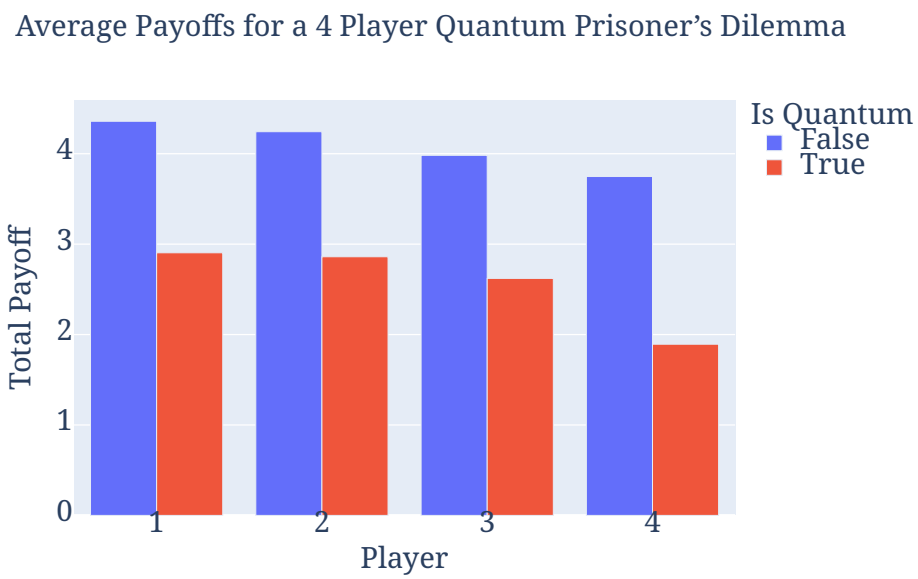


Figure 4.16: Average Payoff per Player for Quantum Knowledge Sharing using CH

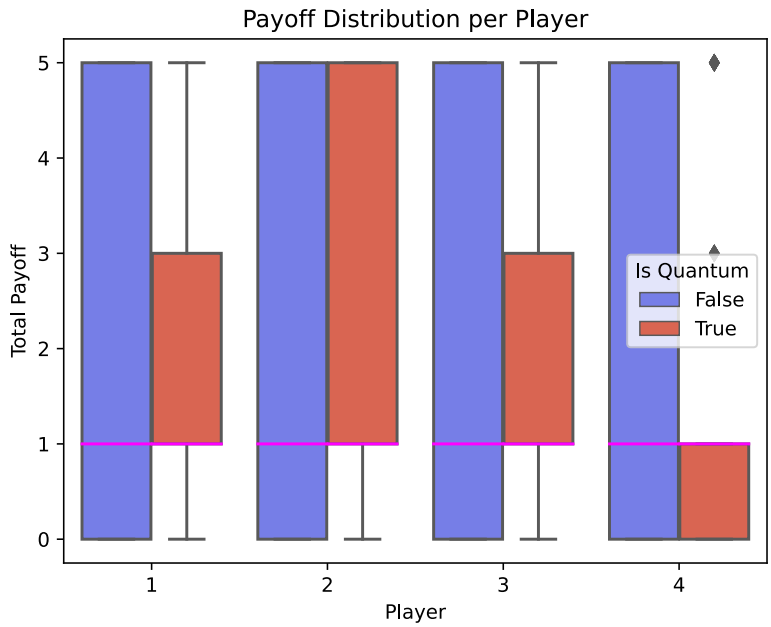


Figure 4.17: Payoff per Player and Quantum Player Position

5. Effects of different quantum strategies

The last case we examined was how the quantum players would behave when using different quantum strategies. Quantum players using the Hadamard gate consistently produce better results that their classical counterparts. To further analyze the effects of the different quantum gate the quantum players may use, we experimented by using the Pauli-Z Gate (Z) 4.2. The Z gate applies a rotation of π radians around the z-axis of the qubit, essentially mapping $|1\rangle$ to $|-1\rangle$.

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{4.2}$$

5..1 Experiments on Aer Simulator

Figure 4.14 shows the average payoff per player for both their classical and quantum iterations in a game with the quantum gate Z as the quantum player strategy. In this case, a quite interesting pattern forms. Most quantum players retain their improved results, except for specifically player 1, whose payoffs significantly worsen, being outperformed than even classical players.

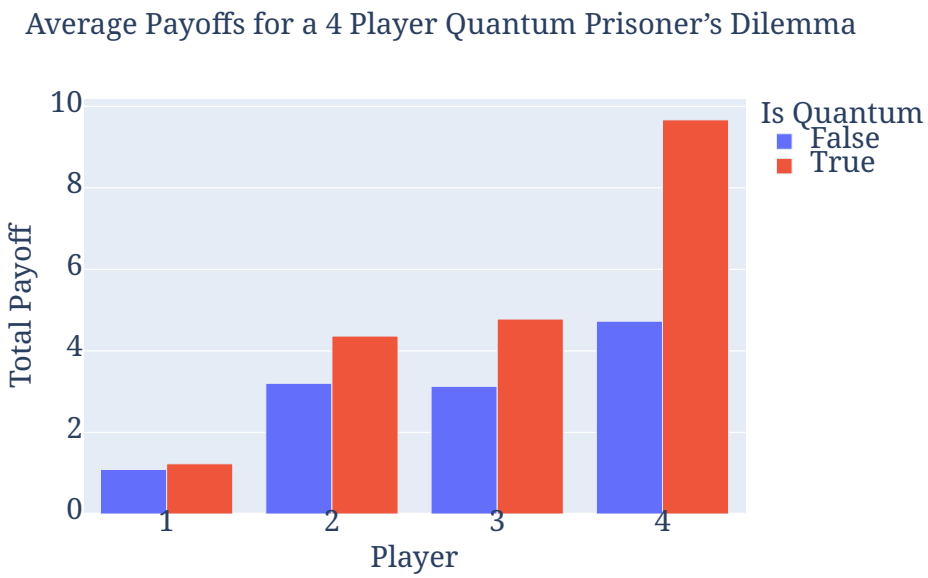


Figure 4.18: Average Payoff per Player and Quantum Player Position

As this behavior is not aligned with our expectations from the theory, we proceeded with analyzing the results further. Upon examining the results of each run,

we came upon an exceptionally interesting conclusion, For all possible graph formulations, the simulator returned exactly one state, with a probability of 1.0. This signifies that this implementation leads to a deterministic result. The quantum states returned are as follows (note the states are in reverse ordering, with the least significant bit being the left-most one, for the convenience of the reader):

State	Probability
10000000	1.0
10101000	1.0
00001010	1.0
00100010	1.0

Figure 4.19: Returned states for quantum strategy Z

These results showcase an interesting pattern; when using the Z quantum gate for the quantum strategy, it leads to games where the left-most bit in the game always wins. For example, in the case of state $|10000000\rangle$, we have a combination of 4 games represented by the states $|10\rangle, |00\rangle, |00\rangle, |00\rangle,$. Similarly, for the state $|10101000\rangle$, we have the states $|10\rangle, |10\rangle, |10\rangle, |00\rangle$. For all cases, every game affected by a quantum player (or by a player who encountered a quantum player), the result is always $|10\rangle$, with the player represented by the left most qubit defecting, and the other cooperating. As in our formulation, the order in which the players play is static, this leads to games where the first player always wins, and the second always loses. By recalling Figure 3.3, we can validate our results by noting the position of each player in their participated games. For example, as player 1 always competes as the left-most qubit (Games 1 and 2), they have the lowest possible payoff. Players 2 and 3 play once as the left-most and once as the right-most qubits, and similarly their results are a bit worse than player 1. Player 4 plays in both games as the right-most bit, which is reflected in their total payoff of 10, meaning in both games they participated in they cooperated while the opponent defected.

The payoff distribution of each player is also similarly affected, with both classical and quantum counterparts of player 1 having outstanding performance with all their payoffs in the range of $[0,1]$, followed by players 2 and 3 with a mixed performance showcasing payoffs between $[1,3]$ and lastly player 4 who again displays a payoff of exactly 5 in every round they participated in. This is in line with our calculations as all games will lead to the state $|10\rangle$, meaning that the first player always Defects, while the opponent always Cooperates.

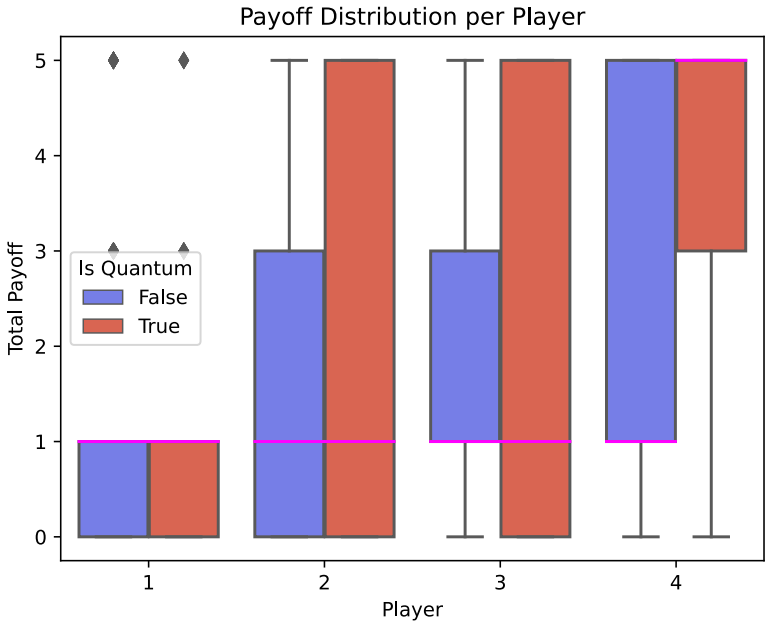


Figure 4.20: Payoff per Player and Quantum Player Position

5..2 Experiments on QPU

Similarly to the experiments conducted with the Aer Simulator, the bit ordering of each game affects the result, with the players on the left most bit always achieving better payoffs than players in the right most one. The slight variations in the exact average payoffs for all players can be attributed to the result of noise when using a QPU. The payoff distributions of all players showcase a lot more variance, again attributed to the noise present in real quantum systems, with a small distinction for players 2 and 3 who are able to achieve a median payoff of 1 instead of the median in Aer's simulator of 3.

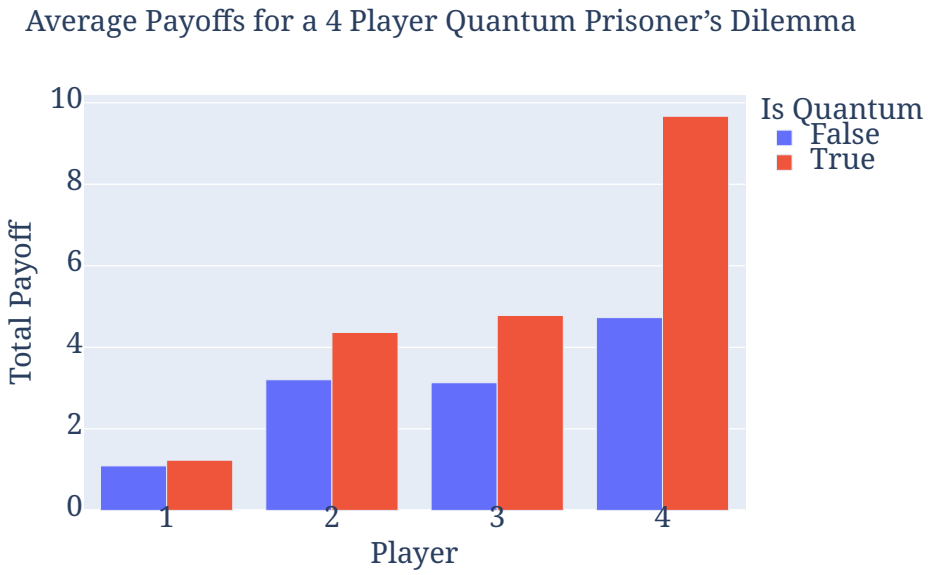


Figure 4.21: Average Payoff per Player and Quantum Player Position

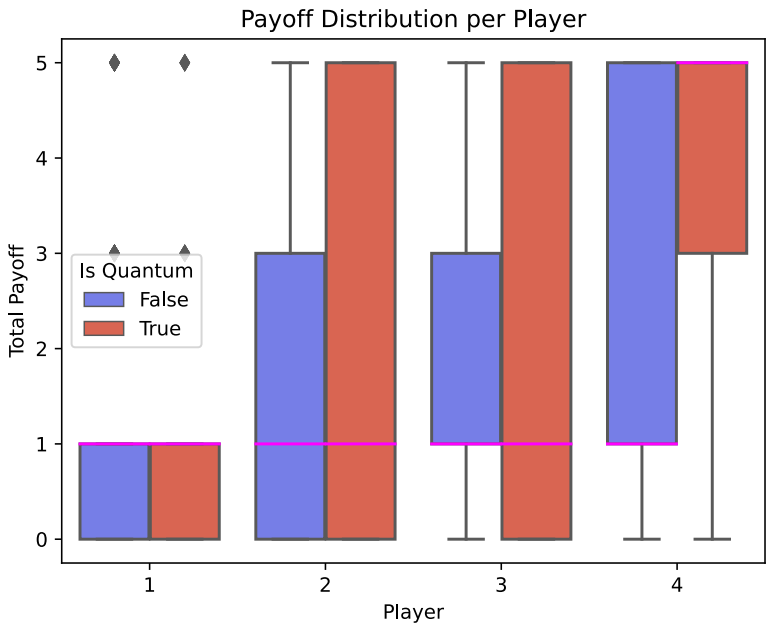


Figure 4.22: Payoff per Player and Quantum Player Position

Chapter 5

Conclusions

In this thesis, we have considered various implementations, computational variations, and comparisons of different compositions for the Quantum Prisoner's Dilemma game on networks. We proposed a technique for encoding a network of quantum games to a quantum circuit and an approach for creating a memory system, where players remember their last choice. We performed several experiments both on simulated quantum environments using IBM's Aer simulator backend, and real quantum systems using IBM's QPU backend and came to the following conclusions:

- In most cases quantum players exhibit a significant advantage against classical players.
- The implementation of knowledge sharing acts as a way to "even the playing field" between classical and quantum players, as it provides classical players with access to quantum strategies on their second round of playing.
- Because of the aforementioned knowledge sharing, the position of a player in the network, and subsequently the order in which the games are played heavily affects one's results, favoring later players.
- Different implementations of knowledge sharing affect the results of the whole game, with the CH gate outperforming the $CNOT$ implementation.
- Different quantum strategies provide significantly different results, with the Z quantum gate making the game completely deterministic.
- For specific cases such as when using the Z gate for the quantum strategy, the bit-order representing the players will be the decisive factor for their payoffs.

The above conclusions are particularly significant, as they both validate the advantage of quantum players in networks of other players, and indicate the possibility of improving these results even further upon further experimentation.

Chapter 6

Future Work

Our computational results indicate that our formulation of quantum prisoner's dilemma on networks is very promising. We intend to continue this line of research and extend its usage to a variety of different implementations. We intend to evaluate experimentally several new configurations, with the major and most interesting ones listed below:

- Increasing the number of players, both quantum and classical in the network to examine the behavior of all players in a more scaled-up system.
- Using different network graphs for the games' setup. Directional graphs may provide an exciting search space by changing the usual flow of the games.
- Experimenting with new quantum strategies and entanglement configurations as shown in [33]
- Incorporating repeating games to examine the players' behavior on a continual system.
- Introducing a system for classical players to be able to select both classical strategies with reasoning.

As the application of Quantum Prisoner's Dilemma on Networks is a novel and less researched area, the possibilities for furthering this thesis are nearly limitless, and the aforementioned suggestions are but a fraction of the viable improvements.

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